Ctrl Alt Biology

Using AI tools for protein design

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Table of content

The Bloom Generative Biology

- 2024 Nobel Prize in Chemistry awarded in part to David Baker for computational protein design.
- AI has enabled the design of entirely new proteins beyond those found in nature.
- Rapid expansion of protein design methods since 2021, including RFdiffusion, ProteinMPNN, BindCraft, AlphaProteo, Chai-1, Boltz-gen …
- Growing scientific and commercial impact, with numerous startups focused on AI-driven protein engineering and drug discovery.

Qiao et al., Nat. Mach. Intell., 2024

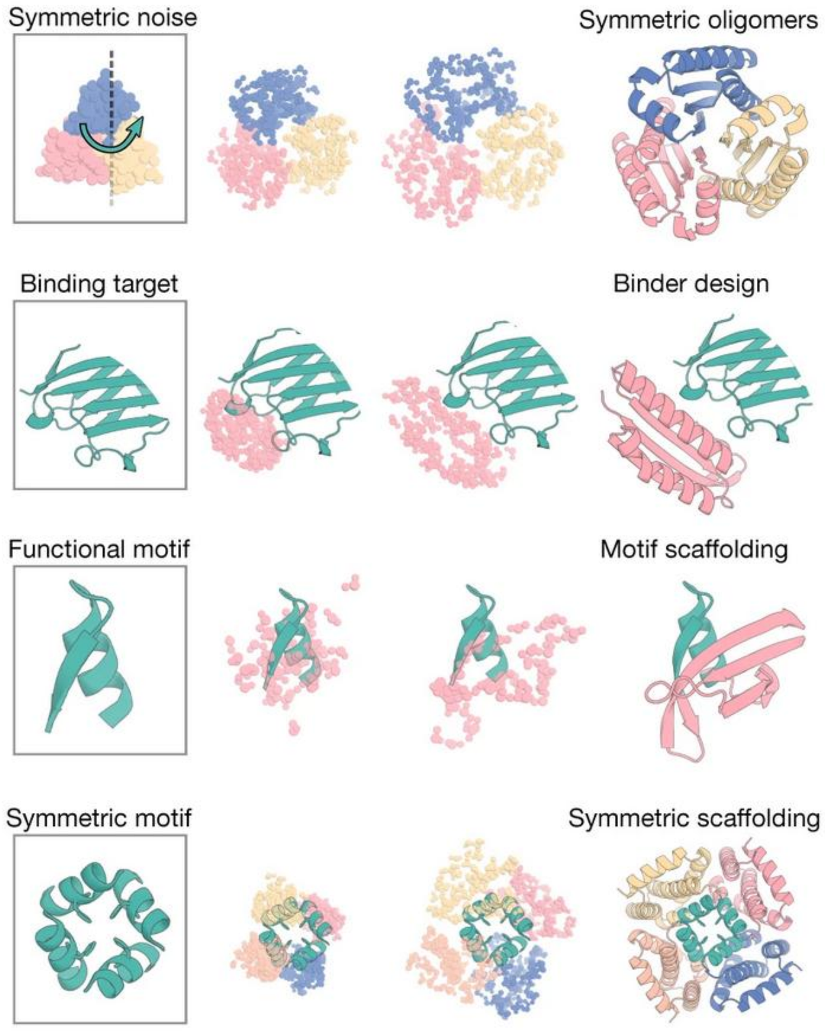
The main pipeline

Protein Design Fundamentals

- Protein sequence determines folding, but the mapping is many-to-one and context-dependent.

- A good design must satisfy several constraints at once:
  - stable fold
  - correct secondary structure
  - packed hydrophobic core
  - compatible surface residues
  - absence of obvious clashes or buried polar atoms
  - desired binding or catalytic geometry, if applicable

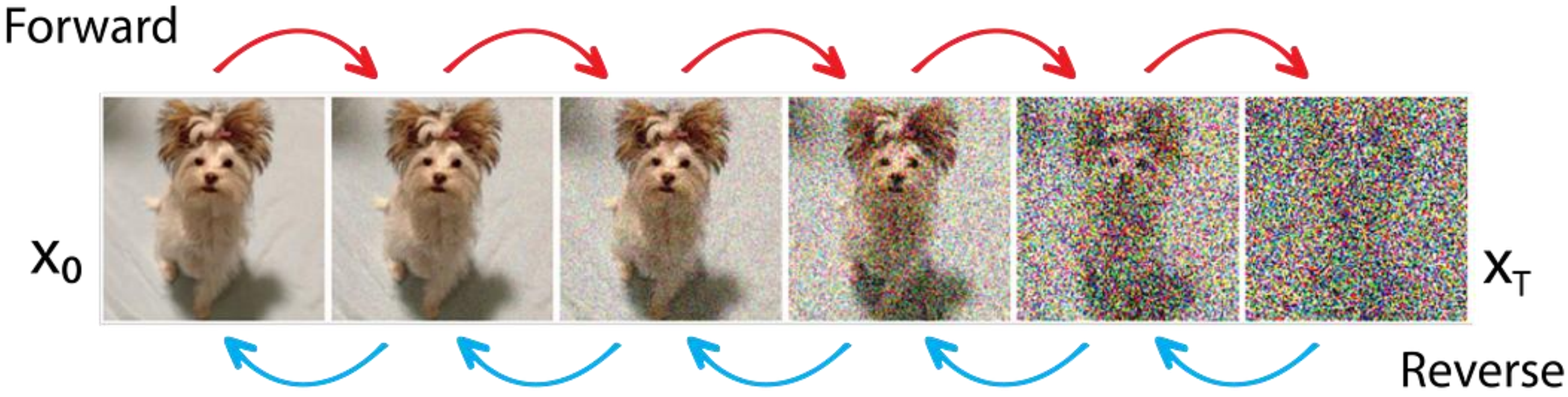
- Separate three design problems:
  - Structure design: what shape should the protein have?
  - Sequence design: what amino acids encode that shape?
  - Function design: what should the structure do?



Watson et al., *Nature* (2023)

Fold Diffusion: Diffusion Models

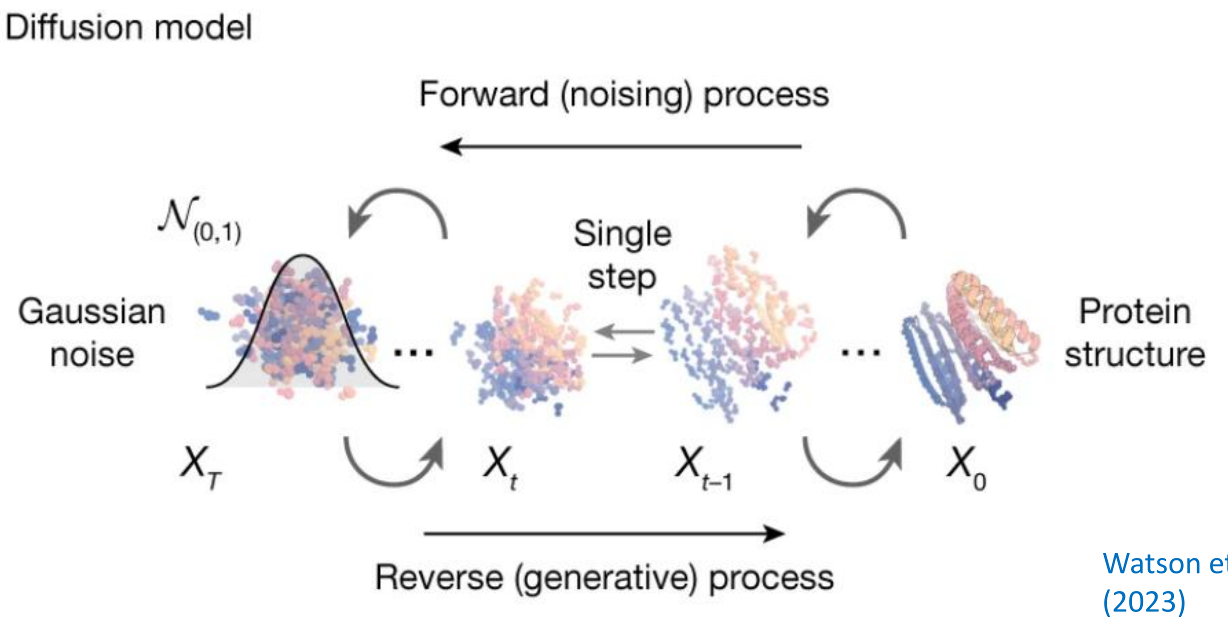
- Generative models that learn to create new data by reversing a gradual noise-adding process.
- Training consists of learning how to transform noisy examples back into realistic samples.
- Generation starts from random noise and iteratively refines it into a valid output.
- Originally developed for image generation, but now applied to proteins, molecules, RNA, and other biological structures.



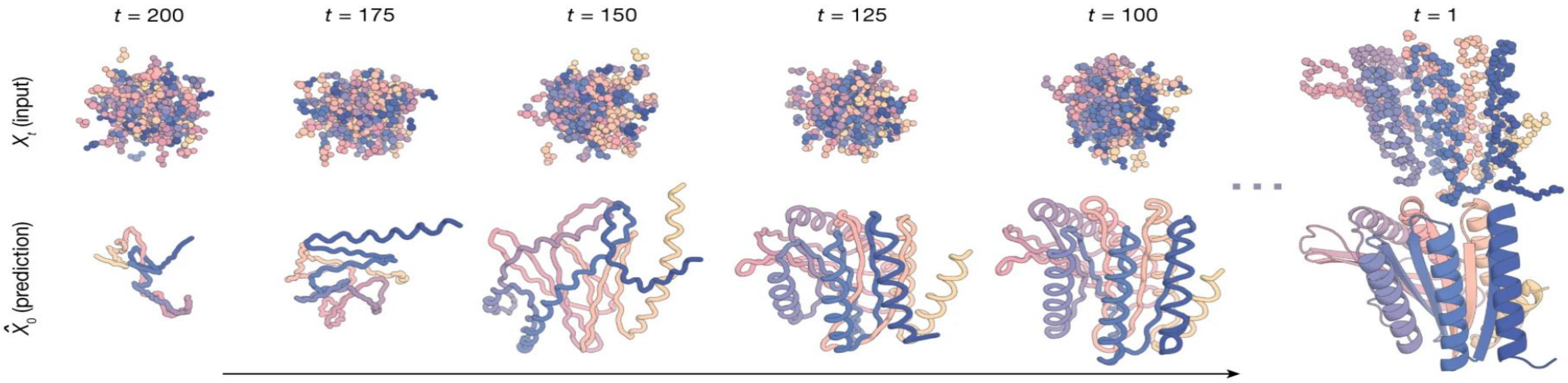
Sayed Ebad, Medium (2024)

Fold Diffusion: Generating Novel Backbones

In protein design, diffusion models generate novel 3D protein backbones that satisfy specified structural or functional constraints.



Watson et al., *Nature* (2023)

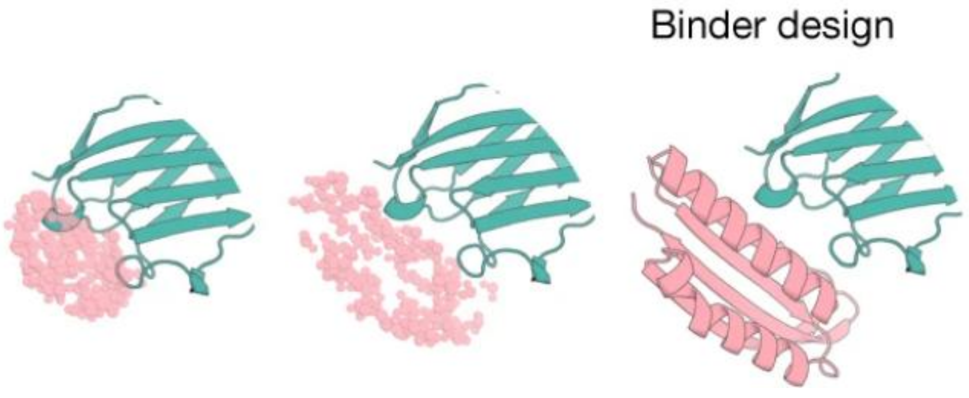


Fold Diffusion: Main differences and Challenges

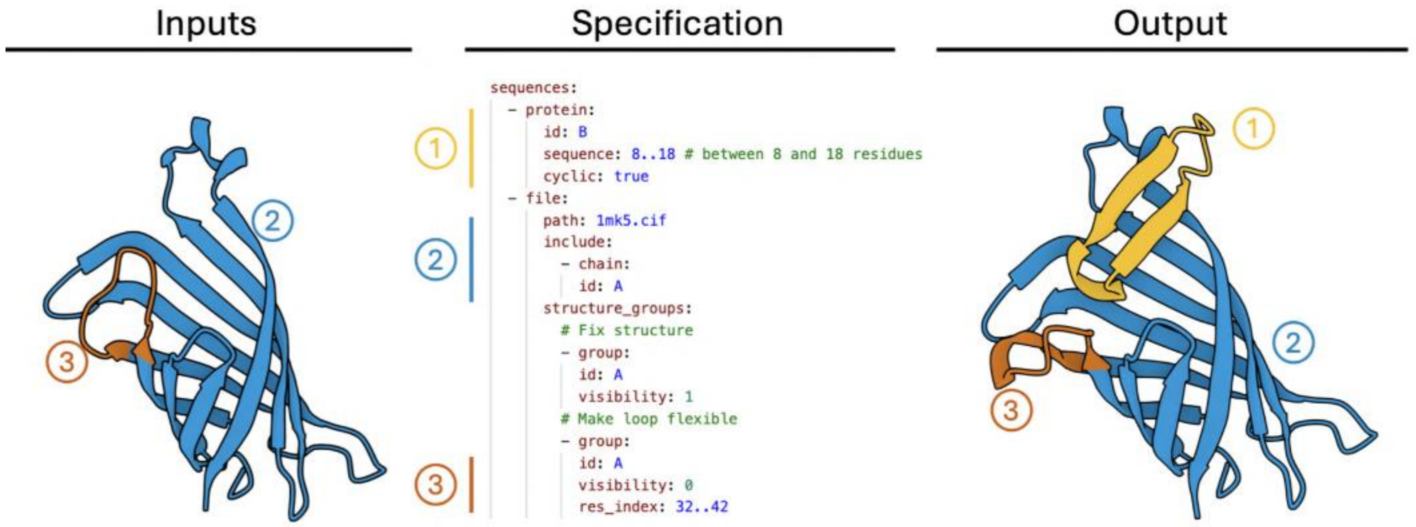
- The primary differences between fold diffusion models lie in the constraints, conditioning strategies, and input modalities they support (e.g., motifs, binding interfaces, symmetry, secondary structure, or target-guided design).

- Finding the right design setup often requires substantial trial-and-error, as small changes in conditioning can lead to very different structural outcomes.

- On a technical level, models differ in their structure representations, encoding schemes, and neural network architectures, which influence design diversity, controllability, and computational cost.



Watson et al., *Nature* (2023)



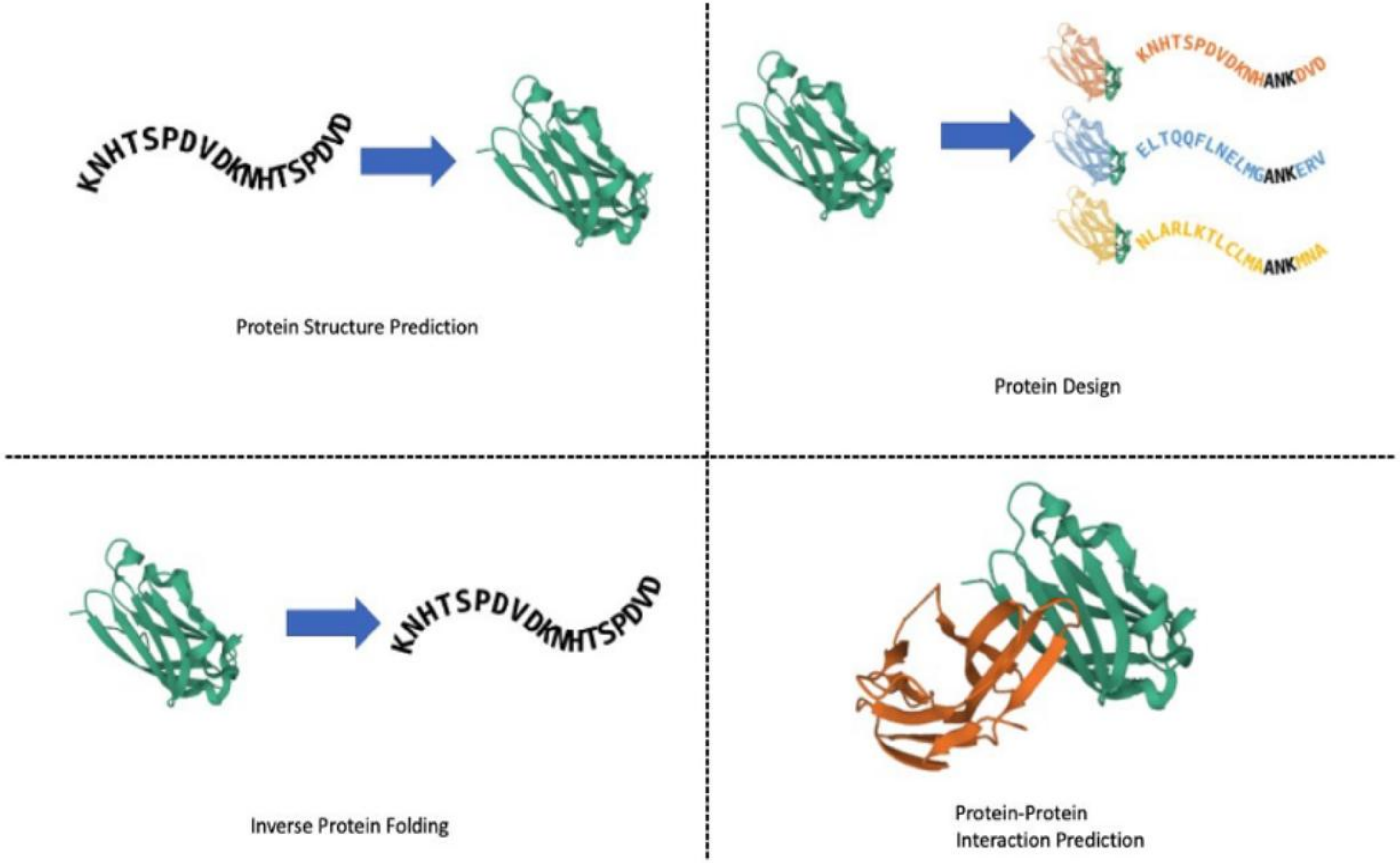
Cho et al., bioRxiv (2025)

Inverse Folding: Designing Sequences for a Backbone

- Inverse folding solves the reverse of structure prediction:
  structure prediction: sequence → structure
  inverse folding: structure → sequence

- Given a target backbone, models such as ProteinMPNN or ESM-IF propose amino acid sequences compatible with that backbone.

- Sequence design usually involves sampling many sequences per backbone, not just one.



Broyde, AWS HPC Blog (2023)

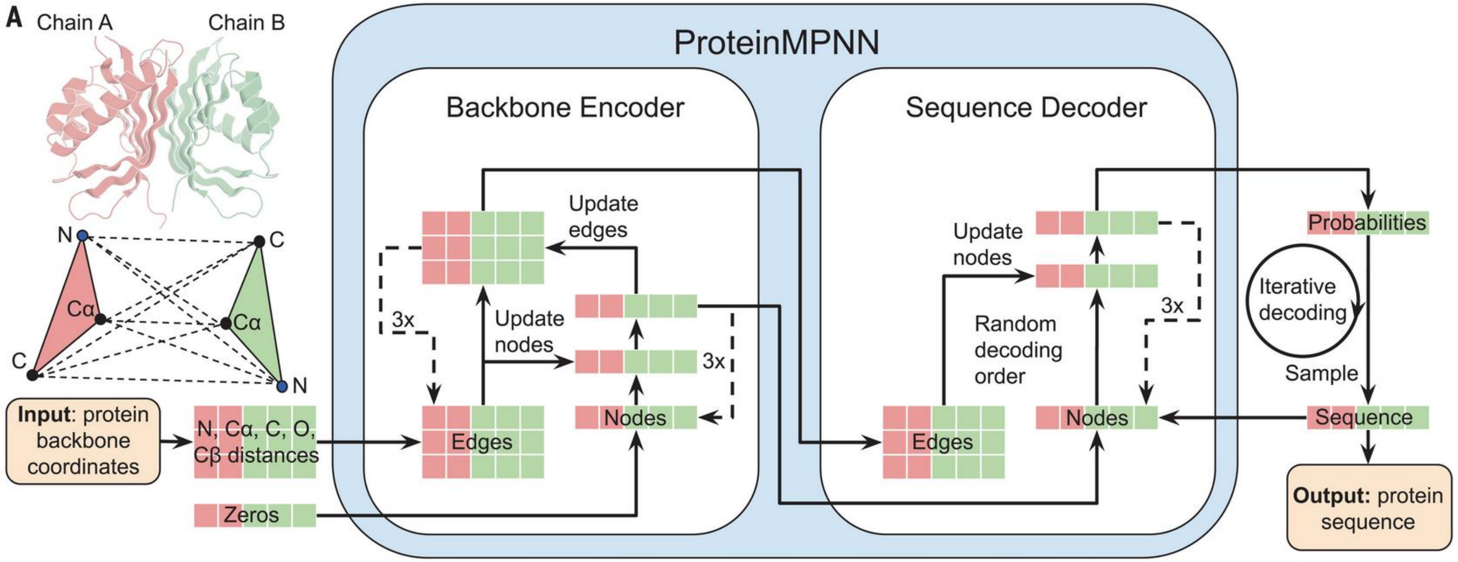
Inverse Folding: Main differences and Challenges

- Inverse folding is typically the stage where the largest number of candidate sequences are generated for a given backbone.

- Since sequences are sampled from a learned probability distribution, models may produce designs that satisfy structural constraints but are undesirable from an experimental perspective.

- Models differ substantially in their architectures, sequence representations, and conditioning strategies.

- Potential liabilities:
  - Unwanted post-translational modification sites (e.g., glycosylation),
  - Accidental disulfide bonds
  - Aggregation-prone regions,



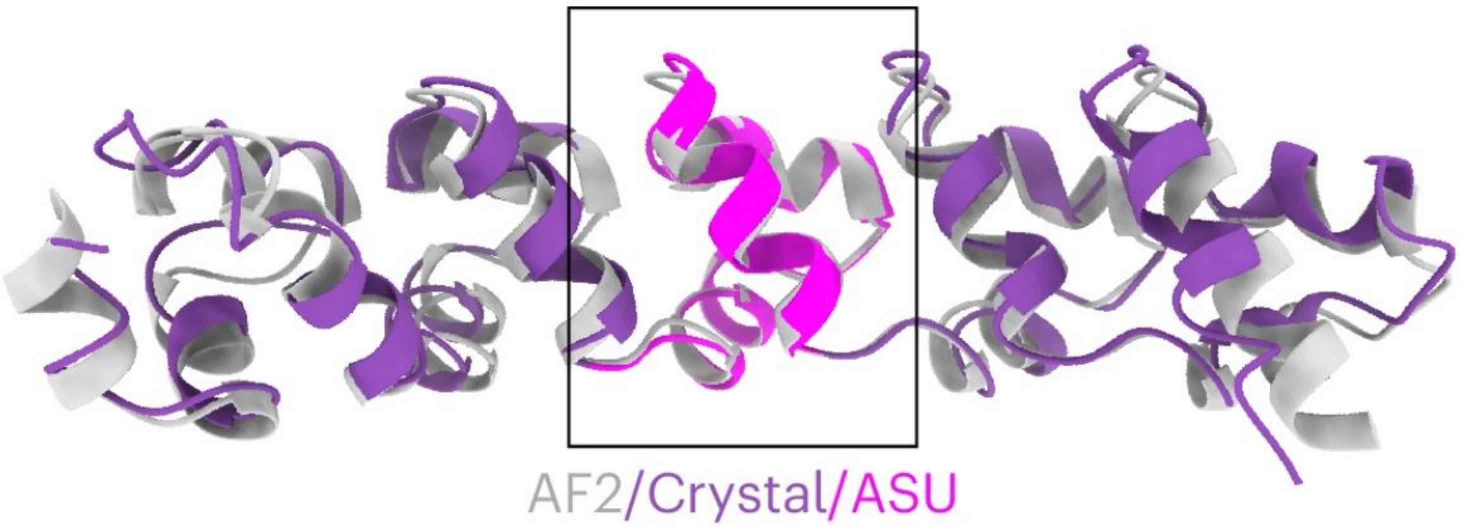
Dauparas et al., *Science* (2022)

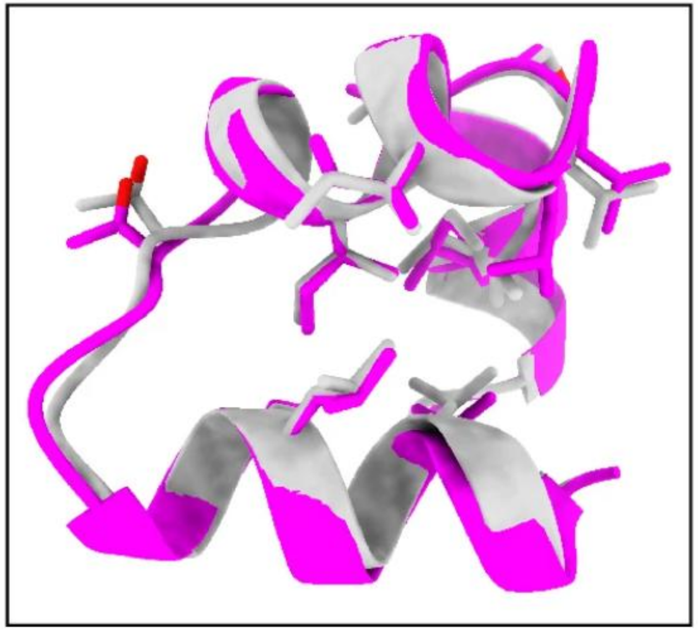
Refolding: Checking the Sequence

- After sequence design, predict the structure of the designed sequence independently.

- The key question is: **Does the sequence fold back to the intended backbone?**

- Compare the predicted/refolded structure against the design target.
  - backbone RMSD or TM-score to the target
  - secondary-structure agreement
  - whether functional motifs remain in place
  - whether interfaces are preserved
  - whether flexible or low-confidence regions appear





Lisanza et al., Nat. Biotechnol. (2025)

Refolding: Main differences and Challenges

Folding methods differ in both input modalities and overall architecture.

Key points:

- High confidence does not imply a good design.
- A sequence may confidently fold into the wrong structure.
- Structural prediction does not assess:
  - Expression
  - Solubility
  - Aggregation propensity
  - Thermostability
  - Function

- **Successful refolding is necessary but not sufficient criterion for selecting designs**.

- Most folding models do an approximate treatment of sidechains placement.

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Validation and Filtering: Computational and Experimental

- Prediction models to assess whether the designed sequence has the desired properties.
- Molecular dynamics simulations evaluate structural stability, flexibility, and persistence of key interactions over time.
- Biophysical scoring tools (Rosetta, FoldX, EvoEF2, etc.) estimate properties such as folding energy, interface quality, packing, and mutation effects.
- Experimental validation remains the ultimate benchmark, including expression, purification, stability measurements, binding assays, and structural characterization.
- Experimental results can be used to guide new generation or improve models

Limitations, Challenges, and Future Directions

Limitations and Challenges

- Many important properties remain difficult to control (expression, solubility, stability, specificity, function).
- Available constraints are often insufficient, requiring significant trial-and-error.
- Binder design works best for rigid targets; flexible regions and conformational changes remain challenging.
- High computational scores do not guarantee experimental success.

Interpreting New Methods

- New design methods are released continuously.
- Most studies claim improved performance, but standardized benchmarks are lacking.
- Differences in datasets, filtering, and success metrics complicate comparisons.
- Benchmark methods on your own design problems whenever possible.

Future Directions

- Increasing integration of backbone generation, sequence design structure prediction and prediction (validation) into unified models.

Take Home Message

- Modern protein design follows a simple workflow: Fold diffusion → Inverse Folding → Refold → Validate.
- Success depends on the combination of methods and filters, not on any single model or metric.
- Be critical when evaluating new approaches: benchmarks are often difficult to compare and may not translate to your application.
- Whenever possible, validate with known data.